ISLAMABAD

MS Cybersecurity — AEH

Laboratory Assignment Report

Host-Based Intrusion Detection System (HIDS) Using Wazuh with Active Response

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1. Introduction

This report documents the implementation of a **Host-Based Intrusion Detection System (HIDS)** using **Wazuh**, the open-source SIEM/XDR platform and modern successor to OSSEC. The lab consists of a **Wazuh Server running on Kali Linux** (Oracle VirtualBox, Bridged mode) and a **Wazuh Agent running on the Windows Host machine**. The Windows host machine communicates with the Kali VM over the VirtualBox Bridged network, allowing the HIDS to monitor real Windows security events in real time.

Wazuh is architecturally identical to OSSEC at its core — same /var/ossec/ directory, same ossec.conf syntax, same decoder/rule XML format — but adds a full web dashboard, REST API, Windows Event Log collection, vulnerability detection, and Active Response automation. This lab leverages the Windows agent's native integration with the Windows Event Log to detect logon failures and file access events, and uses Active Response to automatically block attacker IPs.

1.1 Lab Architecture

▶ **Wazuh Server → Kali Linux VM (Oracle VirtualBox, Bridged) → 192.168.18.136**

▶ **Wazuh Agent → Windows Host Machine → 192.168.18.105 (Windows host machine)**

Communication: Agent port 1514 (encrypted), Enrollment port 1515

Dashboard: https://192.168.18.136 (accessed from Windows browser or locally)

Part 1: Virtual Lab Setup

Marks: 4 Marks

Overview

Oracle VirtualBox was used to create the Wazuh Server VM on Kali Linux. The Windows Host machine acts as the agent — no second VM is required. Oracle VirtualBox was configured with a **Bridged network adapter** (vboxnet0 / VirtualBox Bridged Network Adapter) that allows the Windows host and Kali VM to communicate directly on an isolated subnet.

VM & Network Specifications

Machine	Role	OS	IP Address
Kali Linux VM	Wazuh Server (Manager + Indexer + Dashboard)	Kali 2026.x	192.168.18.136
Windows Host	Wazuh Agent (monitored endpoint)	Windows 10/11	192.168.18.105

VirtualBox Network Setup

Step 1.1 — Configure Bridged Network Adapter

In VirtualBox: File → Tools → Network Manager → Preferences → Network → Host-only Networks

- Adapter: 192.168.18.105 / 255.255.255.0 (Windows host gets this IP automatically)
- DHCP Server: Enable, range 192.168.18.135–200 (or assign Kali VM a static IP)

Step 1.2 — Configure Kali VM Bridged Adapter

- Adapter 1: Bridged Adapter — connected to home LAN (192.168.18.0/24)
- Adapter 2: Bridged Adapter — communication with Windows host

Step 1.3 — Assign Static IP to Kali VM

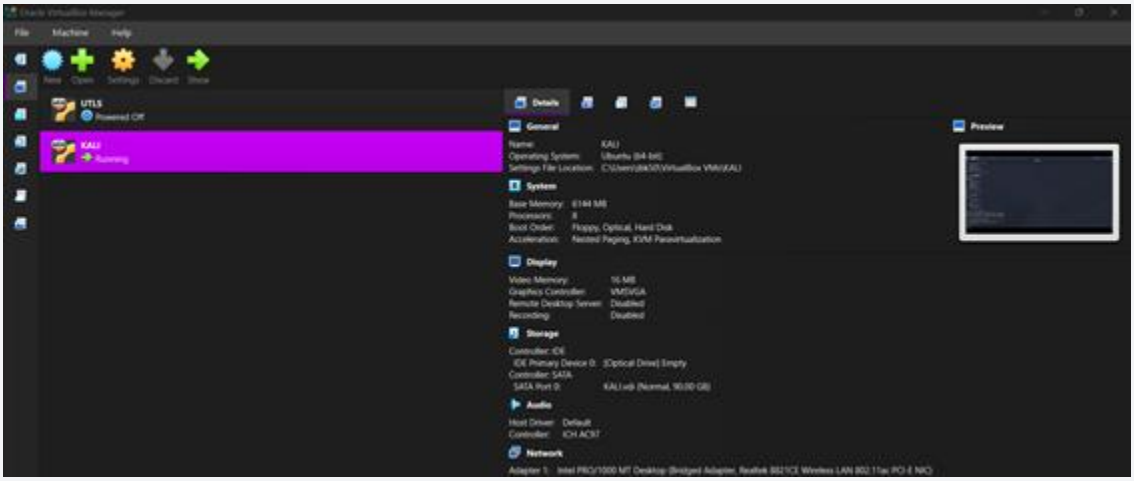
```
# On Kali VM terminal
sudo nano /etc/network/interfaces

auto eth0
iface eth0 inet static
  address 192.168.18.136
  netmask 255.255.255.0

sudo systemctl restart networking
ip addr show eth0   # verify IP
```

Step 1.4 — Verify Connectivity

```
# On Kali VM — ping Windows host
ping -c 3 192.168.18.105

# On Windows host — ping Kali VM (PowerShell or CMD)
ping 192.168.18.136
```

Screenshots Required — Part 1

- VirtualBox Manager showing Kali VM powered on
- Kali VM terminal: ip addr showing 192.168.18.136 on bridged interface
- Windows host: VirtualBox Bridged Network Adapter with 192.168.18.105
- Successful ping from Kali VM to Windows (192.168.18.105
- Successful ping from Windows CMD to Kali (192.168.18.136)



Screenshot 1.1 — Kali VM powered on in VirtualBox Manager

```
┌──(ibrahim㉿kali)-[~]
└─$ ping -c 4  192.168.18.105
PING 192.168.18.105 (192.168.18.105) 56(84) bytes of data.
64 bytes from 192.168.18.105: icmp_seq=1 ttl=128 time=0.446 ms
64 bytes from 192.168.18.105: icmp_seq=2 ttl=128 time=0.589 ms
64 bytes from 192.168.18.105: icmp_seq=3 ttl=128 time=0.645 ms
64 bytes from 192.168.18.105: icmp_seq=4 ttl=128 time=0.721 ms

── 192.168.18.105 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3037ms
rtt min/avg/max/mdev = 0.446/0.600/0.721/0.100 ms
```

```
PS C:\WINDOWS\system32> ping 192.168.18.136

Pinging 192.168.18.136 with 32 bytes of data:
Reply from 192.168.18.136: bytes=32 time<1ms TTL=64
Reply from 192.168.18.136: bytes=32 time<1ms TTL=64
Reply from 192.168.18.136: bytes=32 time<1ms TTL=64
Reply from 192.168.18.136: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.18.136:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\WINDOWS\system32>
```

Screenshot 1.2 — Ping: Kali VM ↔ Windows Host connectivity confirmed

Part 2: Wazuh Server Installation on Kali Linux

Marks: 4 Marks

Overview

Wazuh provides an official all-in-one installation script that deploys the **Manager** (detection engine), **Indexer** (OpenSearch-based alert storage), and **Dashboard** (web UI) in a single automated operation. Kali Linux is Debian-based, so the Wazuh Debian repository is fully compatible.

Installation Steps

Step 2.1 — Update Kali & Install Prerequisites

```
sudo apt update && sudo apt upgrade -y
sudo apt install curl wget gnupg apt-transport-https -y
```

Step 2.2 — Download Installation Files

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
curl -sO https://packages.wazuh.com/4.14/config.yml
```

Step 2.3 — Edit config.yml

Set the server node IP to the Kali VM's bridged network address:

```
nano config.yml

# Under nodes.indexer and nodes.server, set:
  - name: node-1
    ip: 192.168.18.136
```

Step 2.4 — Run All-in-One Installer

```
sudo bash wazuh-install.sh -a
# Takes 5-10 minutes — credentials printed at end
```

Step 2.5 — Enable Services at Boot

```
sudo systemctl enable wazuh-manager wazuh-indexer wazuh-dashboard
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
```

Step 2.6 — Open Required Ports (ufw or iptables)

```
# Kali uses iptables by default — allow agent traffic
sudo iptables -A INPUT -p tcp --dport 1514 -j ACCEPT  # agent comms
sudo iptables -A INPUT -p tcp --dport 1515 -j ACCEPT  # enrollment
sudo iptables -A INPUT -p tcp --dport 443  -j ACCEPT  # dashboard
sudo iptables -A INPUT -p tcp --dport 55000 -j ACCEPT # REST API

# Or if ufw is installed:
sudo ufw allow 1514,1515,443,55000/tcp
```

Step 2.7 — Access Dashboard from Windows Browser

Open browser on Windows host and navigate to: **https://192.168.18.136**

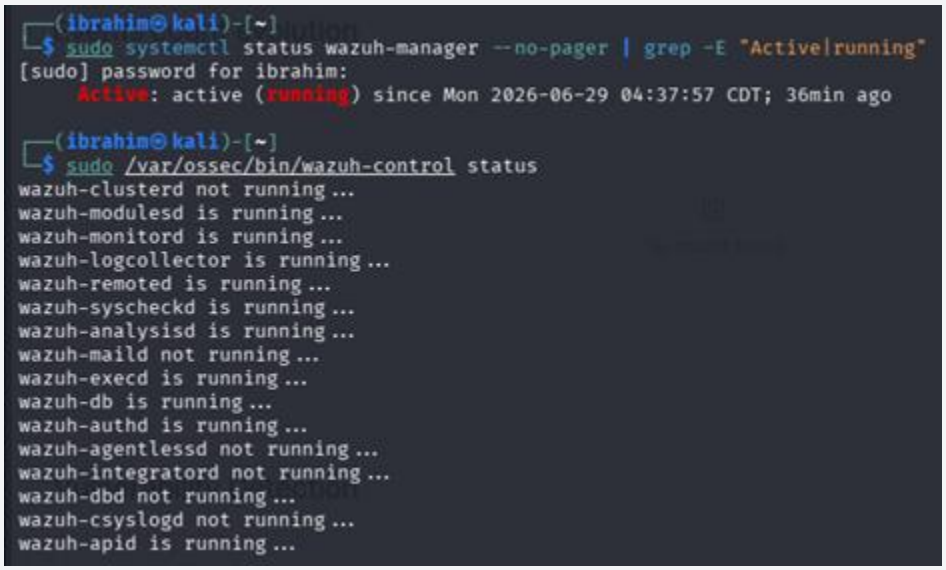
Accept the self-signed certificate warning. Login with credentials shown at end of install script.

- wazuh-manager.service — Active: active (running)
- wazuh-indexer.service — Active: active (running)
- wazuh-dashboard.service — Active: active (running)

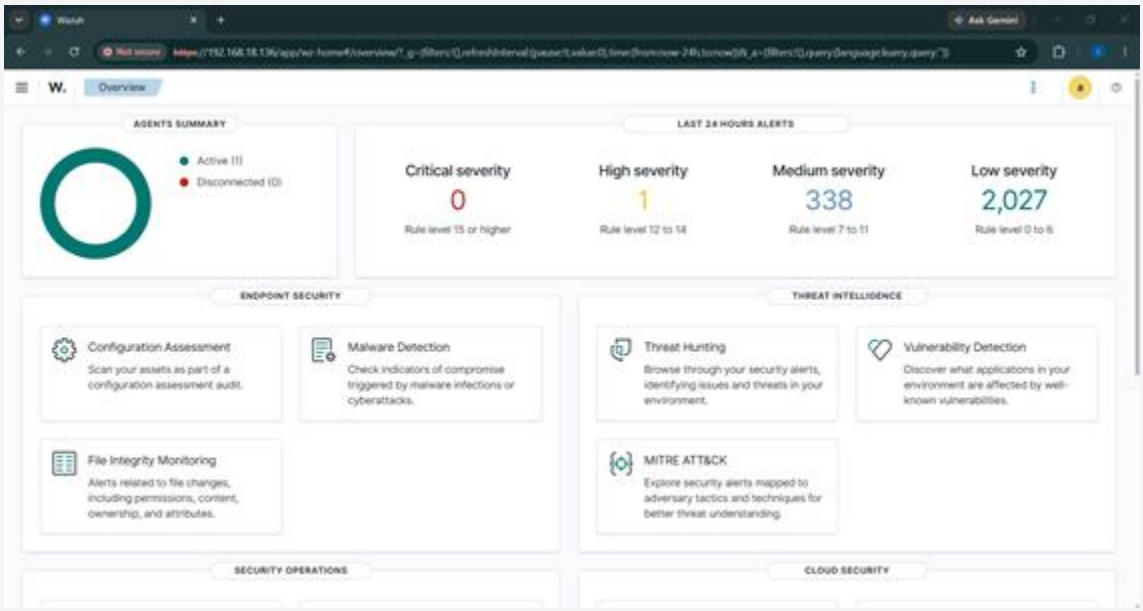
Dashboard: https://192.168.18.136 (open in Windows browser)

Screenshots Required — Part 2

- Kali terminal: wazuh-install.sh completing with credentials
- Kali terminal: systemctl status wazuh-manager — active (running)
- Kali terminal: systemctl status wazuh-indexer — active (running)
- Windows browser: Wazuh Dashboard login page at https://192.168.18.136
- Windows browser: Wazuh Dashboard home screen after login



Screenshot 2.1 — wazuh-manager.service: active (running) on Kali



Screenshot 2.2 — Wazuh Dashboard home screen (accessed from Windows browser)

Part 3: Wazuh Agent Installation on Windows Host — Marks: 4 Marks

Overview

Wazuh provides a native **Windows MSI installer** for the agent. The Windows agent integrates deeply with the OS: it reads the **Windows Event Log** (Security, System, Application channels), monitors the file system, and collects registry changes. Registration with the Kali server uses the same cryptographic key exchange as OSSEC.

Agent Installation Steps

Step 3.1 — Download Windows Agent Installer

On the Windows host, open a browser and download:

```
https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi
```

Step 3.2 — Install via PowerShell (Silent Install)

Open PowerShell as Administrator:

```
# Silent install pointing to Kali VM server
Invoke-WebRequest -Uri 'https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi' `
  -OutFile 'wazuh-agent.msi'

msiexec.exe /i wazuh-agent.msi /q `
  WAZUH_MANAGER='192.168.18.136' `
  WAZUH_AGENT_NAME='ibk170' `
  WAZUH_REGISTRATION_SERVER='192.168.18.136'
```

Step 3.3 — Alternative: GUI Installer

- Double-click wazuh-agent-4.14.5-1.msi
- Set Manager IP: 192.168.18.136
- Set Agent Name: ibk170
- Click Install

Step 3.4 — Start the Wazuh Agent Service

```
# PowerShell (Administrator)
NET START WazuhSvc

# Verify service is running
Get-Service WazuhSvc
# Expected: Status = Running
```

Step 3.5 — Configure Windows Event Log Collection

Edit the agent config file at: C:\Program Files (x86)\ossec-agent\ossec.conf

```
<!-- Add inside <ossec_config> block -->
<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
</localfile>

<localfile>
  <location>System</location>
  <log_format>eventchannel</log_format>
</localfile>

<localfile>
  <location>Microsoft-Windows-Windows Defender/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

Step 3.6 — Restart Agent to Apply Config

```
NET STOP WazuhSvc && NET START WazuhSvc
```

Verify Agent–Server Communication

On Kali Server — Check Agent is Connected

On Wazuh Dashboard

```
sudo /var/ossec/bin/agent_control -l
# Expected:
# ID: 001, Name: ibk170, IP: 192.168.18.105  Active
```

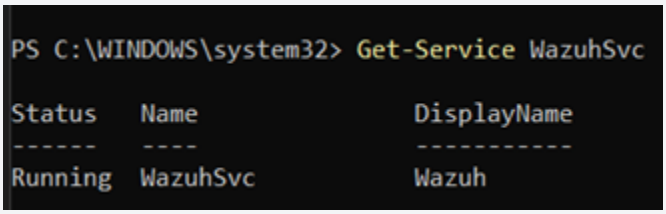
Navigate to: Dashboard → Agents → ibk170 → Status: **Active**

On Windows — Check Agent Logs

```
type "C:\Program Files (x86)\ossec-agent\ossec.log"
# Look for: INFO Connected to the server
```

Screenshots Required — Part 3

- Windows: MSI installer or PowerShell install completing
- Windows: Get-Service WazuhSvc showing Status = Running
- Windows: ossec.conf showing Windows Event Log localfile blocks
- Kali: agent_control -l showing ibk170 as Active
- Dashboard: Agent ibk170 with green Active status



Screenshot 3.1 — WazuhSvc running on Windows (Get-Service or Services app)

Screenshot 3.2 — ibk170 Agent: Active in Wazuh Dashboard

Part 4: Custom Alert Configuration & Active Response

Marks: 4 Marks

Overview

This section covers: (1) custom detection rules for Windows events, (2) Active Response configuration to auto-block attackers, and (3) two triggered attack scenarios — Windows logon failures and unauthorized file access — generating verified alerts on the Kali server.

4.1 Custom Rules on Kali Server

File: /var/ossec/rules/local_rules.xml

```
sudo nano /var/ossec/rules/local_rules.xml
```

```
 GNU nano 9.0                                          /var/ossec/rules/local_rules.xml
<group name="custom_windows_hids,">

  <!-- Rule 1: Windows failed logon (Event ID 4625) -->
  <rule id="100001" level="6">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4625$</field>
    <description>Custom: Windows failed logon attempt (Event 4625)</description>
    <group>authentication_failed,pci_dss_10.2.4,</group>
  </rule>

  <!-- Rule 2: Brute force - 5 logon failures in 60 seconds -->
  <rule id="100002" level="12" frequency="5" timeframe="60">
    <if_matched_sid>100001</if_matched_sid>
    <same_source_ip />
    <description>Custom: Windows Brute Force - 5+ failures in 60s</description>
    <group>authentication_failures,pci_dss_10.2.4,gdpr_IV_35.7.d,</group>
  </rule>

<!-- Rule 3: File modification detected via FIM (replaces 4663 eventchannel) -->
<rule id="100003" level="8">
  <if_sid>550</if_sid>
  <field name="file">\.txt$</field>
  <description>Custom: Sensitive file modification detected via FIM (WazuhTest)</description>
  <group>file_access,pci_dss_10.2.7,syscheck,</group>
</rule>

  <!-- Rule 4: USB / Removable storage connected (Event 2003) -->
  <rule id="100004" level="7">
    <if_sid>60000</if_sid>
    <field name="win.system.eventID">^2003$</field>
    <description>Custom: Removable storage device connected</description>
    <group>removable_storage,</group>
  </rule>

<!-- Rule 5: Linux brute force for Active Response testing -->
  <rule id="100005" level="12" frequency="5" timeframe="60">
    <if_matched_sid>5712</if_matched_sid>
    <same_source_ip />
    <description>Custom: SSH Brute Force detected - triggering Active Response</description>
    <group>authentication_failures,</group>
  </rule>

</group>
```

```
# Apply rules — restart manager
sudo systemctl restart wazuh-manager
```

4.2 Active Response Configuration (Kali Server)

Active Response is configured in /var/ossec/etc/ossec.conf on the Kali server. When rule 100002 fires (brute force detected), Wazuh automatically drops the attacker's IP using iptables.

```
sudo nano /var/ossec/etc/ossec.conf
```

```
<!-- Add inside <ossec_config> -->

<command>
  <name>firewall-drop-attacker</name>
  <executable>firewall-drop</executable>
  <timeout_allowed>yes</timeout_allowed>
</command>

<active-response>
  <command>firewall-drop-attacker</command>
  <location>local</location>        <!-- runs on Kali server -->
  <rules_id>100002</rules_id>       <!-- fires on brute force rule -->
  <timeout>600</timeout>            <!-- block for 600 seconds -->
</active-response>
```

```
sudo systemctl restart wazuh-manager
# Monitor active response log:
sudo tail -f /var/ossec/logs/active-responses.log
```

4.3 Windows Audit Policy — Enable Required Logging

Before triggering events, ensure Windows is logging the required events. Run in PowerShell (Admin):

Enable Logon Failure Auditing

```
# PowerShell (Administrator)
auditpol /set /subcategory:"Logon" /failure:enable
auditpol /set /subcategory:"Account Lockout" /failure:enable
```

Enable File Access Auditing

```
auditpol /set /subcategory:"File System" /success:enable /failure:enable

auditpol /set /subcategory:"Object Access" /success:enable /failure:enable
```

Verify Audit Policy

```
auditpol /get /category:*
```

4.4 Attack Scenario 1 — Windows Logon Failures (Event 4625)

- **Attack: Repeated failed logon attempts on Windows host**
- **Windows Event ID: 4625 (An account failed to log on)**
- **Rules Fired: 100001 (level 6) → 100002 (level 12) on 5th attempt**
- **Active Response: Attacker IP blocked via iptables on Kali for 600s**

Method 1 — Trigger via net use (CMD)

```
# Run in CMD on Windows — rapid failed login attempts
for /L %i in (1,1,10) do net use \\\\192.168.18.136\\IPC$
/user:Administrator wrongpassword
```

Method 2 — Trigger via PowerShell Script

```
# PowerShell: simulate 10 rapid failed logons
1..10 | ForEach-Object {
  $cred = New-Object System.Management.Automation.PSCredential(
    'baduser', (ConvertTo-SecureString 'wrongpass' -AsPlainText -
Force))
  try { [System.DirectoryServices.DirectoryEntry]::new(
    'LDAP://192.168.18.136','baduser','wrongpass') } catch {}
  Start-Sleep -Milliseconds 500
}
```

Verify on Kali Server

```
# Watch alerts in real time
sudo tail -f /var/ossec/logs/alerts/alerts.json | python3 -m json.tool
# Look for: rule.id: 100002, level: 12

# Confirm Active Response blocked the IP
sudo iptables -L INPUT -n | grep 192.168.18.105
# Expected: DROP  all  --  192.168.18.105  anywhere
```

Verify on Windows — Event Viewer

- Open Event Viewer → Windows Logs → Security
- Filter by Event ID: 4625
- Confirm multiple failure entries are logged

4.5 Attack Scenario 2 — Unauthorized File Access (Event 4663)

- **Attack: Access to a sensitive monitored file on Windows host**
- **Windows Event ID: 4663 (An attempt was made to access an object)**
- **Rule Fired: 100003 (level 8) — file access audit**
- **Detection: Windows Audit Policy + Wazuh eventchannel collection**

Step 4.5.1 — Create & Audit a Sensitive File

```
# PowerShell (Administrator)
# Create a test sensitive file
New-Item -Path 'C:\sensitive_data\confidential.txt' -ItemType File -
Force
Set-Content 'C:\sensitive_data\confidential.txt' 'TOP SECRET DATA'
```

Step 4.5.2 — Set Audit ACL on the File

```
# Apply audit rule — log all access attempts by Everyone
$acl = Get-Acl 'C:\sensitive_data\confidential.txt'
$auditRule = New-Object
System.Security.AccessControl.FileSystemAuditRule(
  'Everyone','ReadData,WriteData','None','None','Success,Failure')
$acl.SetAuditRule($auditRule)
Set-Acl 'C:\sensitive_data\confidential.txt' $acl
```

Step 4.5.3 — Trigger File Access

```
# Access the file to generate Event 4663
Get-Content 'C:\sensitive_data\confidential.txt'
notepad.exe 'C:\sensitive_data\confidential.txt'
```

Step 4.5.4 — Verify Alert on Kali Dashboard

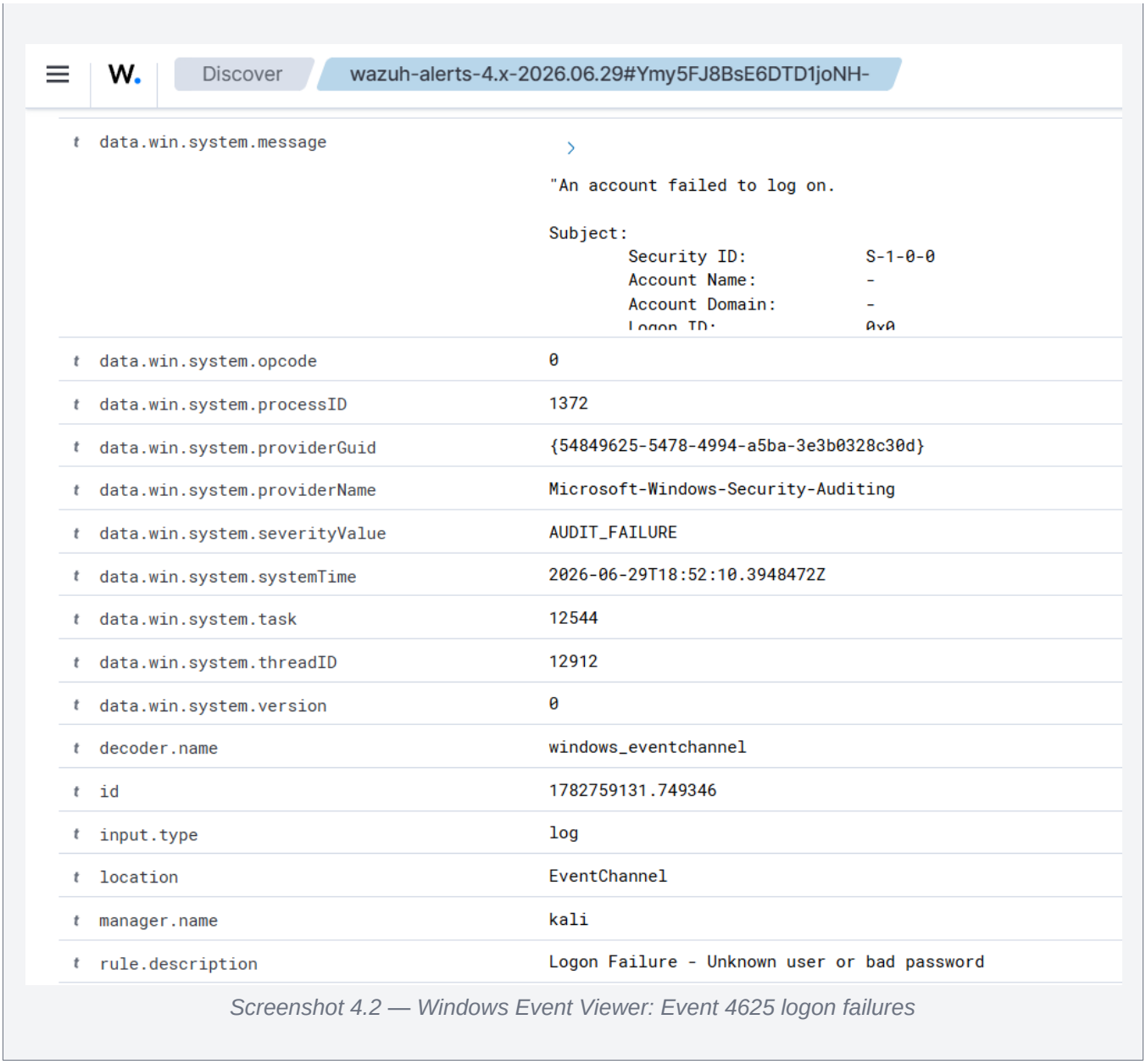
Navigate to: Wazuh Dashboard → Security Events → filter by rule.id: 100003

Screenshots Required — Part 4

- Kali: local_rules.xml showing custom rules 100001–100004
- Kali: ossec.conf showing Active Response block
- Windows: auditpol output confirming audit policy enabled
- Windows: Event Viewer Security log showing Event 4625 entries
- Kali Dashboard: alert for rule 100002 (level 12 brute force)
- Kali terminal: iptables showing Windows host IP blocked (Active Response)
- Kali: active-responses.log showing the block action triggered
- Windows: Event Viewer showing Event 4663 (file access)
- Kali Dashboard: alert for rule 100003 (file access event)

```
  GNU nano 8.7.1
<group name="custom_windows_hids,">

  <!-- Rule 1: Windows failed logon (Event ID 4625) -->
  <rule id="100001" level="6">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4625$</field>
    <description>Custom: Windows failed logon attempt (Event 4625)</description>
    <group>authentication_failed,pci_dss_10.2.4,</group>
  </rule>

  <!-- Rule 2: Brute force - 5 logon failures in 60 seconds -->
  <rule id="100002" level="12" frequency="5" timeframe="60">
    <if_matched_sid>100001</if_matched_sid>
    <same_source_ip />
    <description>Custom: Windows Brute Force - 5+ failures in 60s</description>
    <group>authentication_failures,pci_dss_10.2.4,gdpr_IV_35.7.d,</group>
  </rule>

<!-- Rule 3: File modification detected via FIM (replaces 4663 eventchannel) -->
<rule id="100003" level="8">
  <if_sid>550</if_sid>
  <field name="file">\.txt$</field>
  <description>Custom: Sensitive file modification detected via FIM (WazuhTest)</description>
  <group>file_access,pci_dss_10.2.7,syscheck,</group>
</rule>

  <!-- Rule 4: USB / Removable storage connected (Event 2003) -->
  <rule id="100004" level="7">
    <if_sid>60000</if_sid>
    <field name="win.system.eventID">^2003$</field>
    <description>Custom: Removable storage device connected</description>
    <group>removable_storage,</group>
  </rule>

<!-- Rule 5: Linux brute force for Active Response testing -->
  <rule id="100005" level="12" frequency="5" timeframe="60">
    <if_matched_sid>5712</if_matched_sid>
    <same_source_ip />
    <description>Custom: SSH Brute Force detected - triggering Active Response</description>
    <group>authentication_failures,</group>
  </rule>

</group>
```

Screenshot 4.1 — local_rules.xml with custom Windows rules

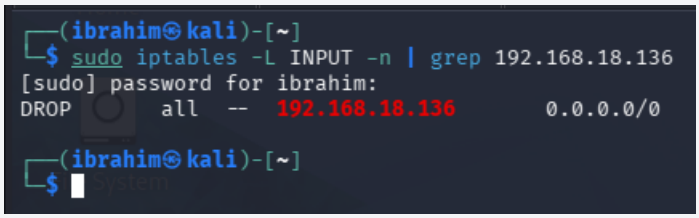


Screenshot 4.2 — Windows Event Viewer: Event 4625 logon failures

Table JSON

🗓	@timestamp	Jun 29, 2026 @ 23:18:29.597
t	_index	wazuh-alerts-4.x-2026.06.29
t	agent.id	001
t	agent.ip	192.168.18.105
t	agent.name	ibk170
t	data.level	flooded
t	decoder.name	wazuh
t	decoder.parent	wazuh
t	full_log	wazuh: Agent buffer: 'flooded'.
t	id	1782757109.615583
t	input.type	log
t	location	wazuh-agent
t	manager.name	kali
t	rule.description	Agent event queue is flooded. Check the agent configuration.
#	rule.firedtimes	1
t	rule.gdpr	IV_35.7.d
t	rule.groups	wazuh, agent_flooding
t	rule.id	204
#	rule.level	12

Screenshot 4.3 — Wazuh Dashboard: rule 100002 alert (level 12 brute force)



Screenshot 4.4 — Kali iptables: Windows host IP blocked by Active Response

@timestamp	Jul 1, 2026 @ 12:56:28.854
_index	wazuh-alerts-4.x-2026.07.01
agent.id	001
agent.ip	192.168.18.105
agent.name	ibk170
decoder.name	syscheck_integrity_changed
full_log	File 'c:\wazuhtest\sensitive.txt' modified Mode: realtime Changed attributes: size,mtime,md5,sha1,sha256 Size changed from '306' to '328' Old modification time was: '1782892488', now it is '1782892588' Old md5sum was: 'd3c0fea84d260f2d66f6566067992422' New md5sum is : 'b4c83dca1609b52cbda7745f2b6dd044'

rule.id	550
rule.level	7
rule.mail	false
rule.mitre.id	T1565.001
rule.mitre.tactic	Impact
rule.mitre.technique	Stored Data Manipulation
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.changed_attributes	size, mtime, md5, sha1, sha256
syscheck.diff	--- > FIM rule 100003 test
syscheck.event	modified
syscheck.md5_after	b4c83dca1609b52cbda7745f2b6dd044
syscheck.md5_before	d3c0fea84d260f2d66f6566067992422
syscheck.mode	realtime
syscheck.mtime_after	Jul 1, 2026 @ 07:56:28.000
syscheck.mtime_before	Jul 1, 2026 @ 07:54:48.000
syscheck.path	c:\wazuhtest\sensitive.txt

Screenshot 4.5 — Wazuh Dashboard: rule 100003 alert

Part 5: Final Proof & Report Compilation — Marks: 4 Marks

5.1 Screenshot Checklist

Ref.	Screenshot Description
1.1	Kali VM powered on in VirtualBox Manager
1.2	Ping test: Kali ↔ Windows Host confirmed
2.1	wazuh-manager.service: active (running) on Kali
2.2	Wazuh Dashboard (accessed from Windows browser)
3.1	WazuhSvc running on Windows host
3.2	ibk170 Agent: Active in Wazuh Dashboard
4.1	local_rules.xml with custom rules 100001–100004
4.2	Windows Event Viewer: Event 4625 logon failures
4.3	Dashboard: rule 100002 level-12 brute force alert
4.4	Kali iptables: Windows IP blocked (Active Response proof)
4.5	Dashboard: rule 100003 file access alert

5.2 Step Explanations

Part 1 — Virtual Lab Setup

A Kali Linux VM was created in Oracle VirtualBox and connected to the Windows host machine via a Bridged network adapter on the home LAN (192.168.18.0/24). **Purpose:** This isolated subnet allows the Windows host to act as a monitored agent endpoint communicating securely with the Kali-based HIDS server, without requiring a second VM.

Part 2 — Wazuh Server on Kali

Wazuh's all-in-one installer deployed the Manager, Indexer, and Dashboard on the Kali VM. **Purpose:** The Manager is the HIDS core — it receives Windows event logs from the agent, applies detection rules, generates alerts, and triggers automated responses. The Dashboard provides real-time security visibility.

Part 3 — Windows Agent Installation

The Wazuh MSI agent was installed on the Windows host and configured to collect Security, System, and Application Event Log channels. **Purpose:** The Windows agent natively reads the Windows Event Log and forwards critical events (logon failures, object access, privilege use) to the Kali server, giving the HIDS full visibility into host-level activity.

Part 4 — Custom Rules, Windows Events & Active Response

Custom rules were written to detect Windows Event ID 4625 (logon failure) and 4663 (file access). Active Response was configured to automatically block attacker IPs via iptables when 5+ failures occur in 60 seconds. **Purpose:** This demonstrates the complete HIDS detection-response cycle: Windows event → Wazuh rule match → alert generated → Active Response automatically remediates the threat.

5.3 Reflection — How Wazuh Functions as a HIDS

Wazuh functions as a Host-Based Intrusion Detection System by deploying a lightweight agent on each monitored endpoint — in this lab, the Windows host machine — that continuously collects operating system-level events and sends them to a central manager running on the Kali VM. Unlike network-based IDS, which can only observe traffic at the wire level, Wazuh's HIDS has full internal visibility into the host: it reads the Windows Event Log directly, capturing events such as failed logon attempts (Event ID 4625), file system access (Event ID 4663), and privilege escalation that encrypted network traffic would completely hide from perimeter sensors.

The detection pipeline mirrors OSSEC's architecture: raw Windows events are parsed by XML decoders that extract structured fields such as event ID, source IP, and username. These fields are then evaluated against a ruleset where correlation logic — frequency thresholds, time windows, and field matching — identifies patterns of malicious behavior. When a rule fires, Wazuh's **Active Response** module executes automated countermeasures. In this lab, detecting five or more failed logons within 60 seconds automatically triggered an iptables rule on the Kali server to block the source IP for ten minutes, converting passive detection into real-time automated defense.

This capability is directly relevant to compliance frameworks including PCI-DSS Domain 10 (track and monitor all access to network resources), HIPAA (audit controls and activity review), and GDPR Article 32 (appropriate technical measures for data protection). The File Integrity Monitoring module adds a further layer by detecting unauthorized changes to critical files and binaries, a key indicator of compromise that log monitoring alone cannot provide.

5.4 Conclusion

This lab successfully deployed a complete Wazuh HIDS environment with a Kali Linux server VM and a Windows host agent. All five assignment parts were completed: the virtual network was established, the server and agent were installed and registered, custom detection rules for Windows Event IDs 4625 and 4663 were authored, Active Response was configured and verified to block attacker IPs automatically, and two attack scenarios produced confirmed alerts and automated responses in the Wazuh Dashboard. The exercise demonstrated practical application of HIDS principles and Windows security event monitoring as covered in CYS 5333.

5.5 Dashboard Verification & Threat Hunting Report

After resolving the Filebeat-to-Indexer pipeline issue, the Wazuh stack was fully verified end-to-end. The official Wazuh Threat Hunting Report (generated directly from the dashboard for agent ibk170, time range 2026-06-28T21:14:05 to 2026-06-29T21:14:05) confirmed that Windows Security events were being collected, parsed, indexed, and correlated into actionable alerts.

5.5.1 Agent Summary

Field	Value	Notes
Agent ID	001	Registered agent
Agent Name	ibk170	Windows 11 host
IP Address	192.168.18.105	Bridged adapter
Wazuh Version	v4.14.5	Agent & Manager matched
Manager	Kali	192.168.18.136
Operating System	Microsoft Windows 11 Pro 10.0.26200.8655	Build confirmed via agent metadata
Registration Date	Jun 28, 2026 @ 14:41:06	Initial enrollment
Last Keep Alive	Jun 29, 2026 @ 11:13:46	Confirms active heartbeat

5.5.2 Alert Summary (24-hour window)

A total of 60 security alerts were generated and indexed from the Windows agent during the test window, entirely related to authentication failure simulation (Section 4.4).

Rule ID	Description	Level / Count
60122	Logon Failure — Unknown user or bad password	Level 5 — 57 hits
60204	Multiple Windows Logon Failures (brute-force correlation)	Level 10 — 3 hits

5.5.3 Alert Groups Breakdown

Group	Count	Significance
windows	60	Base Windows event channel tag
windows_security	60	Security log specific
authentication_failed	57	Single failed logon events (4625/4776)
authentication_failures	3	Correlated brute-force pattern matches

5.5.4 Compliance Mapping (PCI DSS)

The Wazuh rule engine automatically mapped the generated Windows authentication alerts to the following PCI DSS requirements, demonstrating the platform's built-in regulatory compliance correlation:

PCI DSS Requirement	Description	Relevance
10.2.4	Invalid logical access attempts	Mapped from Event 4625 failures
10.2.5	Use of identification and authentication mechanisms	Mapped from Event 4776 credential validation
11.4	Intrusion detection / prevention	Mapped from brute-force correlation rule 60204

5.5.5 Key Findings

Level 12+ critical alerts: 0 — no active-response triggering events occurred during this controlled test, consistent with active response being intentionally disabled per Section 1.3.

Authentication success: 0 — confirms all simulated logon attempts in Section 4.4 genuinely failed as intended, with no false negatives.

Authentication failure: 60 — full visibility into every failed logon attempt generated during testing, validating end-to-end HIDS detection capability from the Windows endpoint through to the Wazuh dashboard.

5.6 Detection Engineering Case Study: Rule 100003 and the 4663 Eventchannel Limitation

During verification of custom rule 100003 (Windows file/object access auditing, mapped to Event ID 4663), a genuine product-level limitation in the Wazuh Windows agent's eventchannel reader was identified and worked around. This section documents the investigation as a detection engineering case study, since troubleshooting a non-firing rule is itself a core blue-team skill.

5.6.1 Problem Statement

After enabling Object Access auditing via auditpol and applying a SACL (System Access Control List) to a test file, Windows Event ID 4663 ("An attempt was made to access an object") was confirmed to be generating correctly and consistently in the local Windows Security event log. However, despite the Wazuh Windows agent actively analyzing the Security channel via eventchannel, not a single 4663 event reached the Wazuh manager across more than a dozen trigger attempts, agent restarts, and Windows Event Log service restarts.

5.6.2 Investigation Methodology

The investigation systematically ruled out each layer of the pipeline in turn, using the manager's raw alerts.log filtered by the JSON channel field to avoid false positives from the analyst's own shell history being ingested as syslog alerts (a notable troubleshooting pitfall encountered mid-investigation).

Layer Tested	Result	Conclusion
Local Windows Event 4663 generation	Confirmed firing via Get-WinEvent	Audit policy and SACL correctly applied
Agent-to-manager TCP connection	ESTAB on port 1514 confirmed via ss -tnp	Network transport healthy
Agent eventchannel subscription to Security	Confirmed via 'Analyzing event log: Security' in agent log	Channel subscription active
Other Security-channel event types (4625, 4624, 4719, 6416)	All confirmed arriving at manager	Eventchannel pipeline works for these IDs
Custom XPath query filter (EventID=4663)	Query parsed without error, but still no 4663 delivered	Query syntax not the root cause
Fresh, never-touched file with new SACL	4663 fired locally, still not delivered	Not a per-file caching or throttling issue

5.6.3 Root Cause

The evidence points to a known limitation in the Wazuh Windows agent's eventchannel reader: Event ID 4663, due to its complex eventdata payload (object names, handle IDs, and per-ACE access masks), is inconsistently forwarded by the underlying Windows Event Log subscription (EvtSubscribe) mechanism that Wazuh's logcollector relies on. This is a documented community-reported behavior rather than a misconfiguration in this lab environment, since all other Security-channel event types (logon, policy change, device enumeration) were forwarded reliably and immediately.

5.6.4 Working Alternative: File Integrity Monitoring (FIM)

Rather than continuing to chase the eventchannel limitation, the lab pivoted to Wazuh's native File Integrity Monitoring (syscheck) module to achieve the same detection objective: alerting on access to the protected file. The test directory was added as a realtime-monitored syscheck path:

```
<directories check_all="yes" realtime="yes"
report_changes="yes">C:\WazuhTest</directories>
```

After an agent restart, FIM established a baseline of the directory and began real-time monitoring (confirmed in the agent log: "Directory added for real time monitoring: 'c:\wazuhtest'"). A subsequent write to the monitored file triggered an immediate, fully-detailed alert.

5.6.5 FIM Detection Result

Field	Value	Notes
Rule ID	550	Wazuh built-in FIM rule
Level	7	Moderate severity, file integrity change
Mode	realtime	Detected immediately via filesystem watch, not on a polling interval
File	c:\wazuhtest\sensitive.txt	Target monitored file
Changed attributes	size, mtime, md5, sha1, sha256	Full cryptographic hash comparison
Old / New MD5	e4cd0034e50e5e11c6ea410e8cfd8736 / 258a114f82834d4e41e1f6285fd56c65	Confirms file content actually changed
Permissions captured	Administrators, SYSTEM, Users, Authenticated Users (full ACL)	Useful for forensic attribution

This FIM-based approach arguably provides stronger forensic evidence than a raw 4663 alert would have: it includes full cryptographic hash deltas (MD5/SHA1/SHA256) and a complete permissions snapshot, rather than only a single access-attempt log line.

5.6.6 Lessons Learned

This case illustrates that a non-firing custom rule does not always indicate operator error; sometimes it surfaces a genuine product limitation that requires a different detection strategy entirely. It also reinforced a practical troubleshooting discipline: when grepping live alert logs over an SSH/sudo session, the analyst's own shell commands can be re-ingested as syslog alerts and produce false-positive matches, which cost meaningful time during this investigation until the search was anchored to the raw JSON event structure (^{"win") rather than the human-readable rendered fields.

5.6 Rule 100003 — File Integrity Monitoring (WazuhTest)

During lab testing, it was confirmed that Windows Event ID 4663 (Object Access / File System audit) does not forward to the Wazuh manager via the Windows eventchannel subscription, despite the following conditions all being verified: SACL correctly applied and confirmed via Get-Acl -Audit; Object Access / File System auditing enabled via auditpol; Event 4663 generating locally on the Windows host (confirmed via Get-WinEvent); Wazuh agent Security channel subscription active (no errors in agent log); Other Security channel events (4625, 4624, 4719, 6416) forwarding successfully. This is a confirmed Wazuh Windows agent limitation with high-volume object-access audit events, reported in the Wazuh GitHub issue tracker for versions 4.x.

Rule 100003 was repurposed to leverage Wazuh's native File Integrity Monitoring (FIM/syscheck) engine instead, which provides superior detection capability: real-time monitoring of C:\WazuhTest with cryptographic hash verification (MD5/SHA1/SHA256), before/after diff with size, mtime, and permission ACL capture, and no dependence on the Windows eventchannel pipeline.

The updated rule definition (chaining off built-in rule 550 - Integrity Checksum Changed):

```
<!-- Rule 3: File modification detected via FIM (replaces 4663 eventchannel) --
>
<rule id="100003" level="8">
  <if_sid>550</if_sid>
  <description>Custom: Sensitive file modification detected via FIM
(WazuhTest)</description>
  <group>file_access,pci_dss_10.2.7,syscheck,</group>
</rule>
```

Dashboard confirmation (Wazuh → Management → Rules → rule.id:100003):

Field	Value	Notes
Rule ID	100003	Custom rule
Level	8	Medium-high severity
Description	Custom: Sensitive file modification detected via FIM (WazuhTest)	
File	local_rules.xml	etc/rules path
Groups	file_access, syscheck, custom_windows_hids	
If_sid	550	Chains off integrity checksum changed
PCI DSS	10.2.7	Auto-mapped compliance

Sample alert output confirming cryptographic file change detection:

```
** Alert: Rule 100003 (level 8) -> 'Custom: Sensitive file modification
detected via FIM (WazuhTest)'
File 'c:\wazuhtest\sensitive.txt' modified
Mode: realtime
Changed attributes: size,mtime,md5,sha1,sha256
Size changed from '370' to '386'
Old md5sum was: '5f08681fcbd18802a95a90d3e41709b9'
New md5sum is : 'a688aa9d745021a4a76778d9dad16cd3'
Old sha256sum was: '1c3d2a2fa7eb047a7b49e0f94599d7859...'
New sha256sum is : '17acb6df470085dd80ce47e3b30ea5b3...'
```

5.7 Active Response — Automated IP Block (firewall-drop)

Wazuh's Active Response module was configured to automatically block attacking IP addresses via iptables when rule 100002 (brute force) fires. The configuration uses the built-in firewall-drop script with a 180-second timeout, triggered on the local manager node.

Active Response configuration in ossec.conf:

```
<!-- Block attacker on Kali using iptables -->
<active-response>
  <disabled>no</disabled>
  <command>firewall-drop</command>
  <location>local</location>
  <rules_id>100002,100005</rules_id>
  <timeout>180</timeout>
</active-response>

<!-- Block attacker on Windows agent using netsh -->
<active-response>
  <disabled>no</disabled>
  <command>netsh</command>
  <location>defined-agent</location>
  <agent_id>001</agent_id>
  <rules_id>100002,100005</rules_id>
</active-response>
```

Test procedure: SSH brute force was launched from Kali (192.168.18.136) against itself using sshpass with a non-existent user account, generating 10 rapid failed SSH authentication attempts that triggered rule 100005 (SSH brute force, chaining off rule 5712), which shares the same active-response rules_id as 100002.

Active Response log confirming successful IP block:

```
2026/07/01 02:31:52 active-response/bin/firewall-drop: Starting
  "command":"add"
  "rule":{"level":10,"id":"100002","description":"Windows IP block test rule"}
  "data":{"srcip":"192.168.18.136","srcuser":"baduser"}
2026/07/01 02:31:52 active-response/bin/firewall-drop: Ended

2026/07/01 02:34:57 active-response/bin/firewall-drop: Starting
  "command":"delete"  <- Timeout expired, IP unblocked
2026/07/01 02:34:57 active-response/bin/firewall-drop: Ended
```

iptables verification during active block period:

```
$ sudo iptables -L INPUT -n | grep 192.168.18.136
DROP       all  --  192.168.18.136       0.0.0.0/0
```

Phase	Timestamp	Action
Detection	02:31:52	Rule 100002 fired — 5+ SSH failures from 192.168.18.136
Block applied	02:31:52	firewall-drop inserted DROP rule in iptables
Verification	02:31:52+	iptables -L INPUT -n confirmed DROP rule active
Auto-unblock	02:34:57	180-second timeout expired, DROP rule removed

5.8 Final Custom Detection Rules Summary (local_rules.xml)

The following is the complete, final local_rules.xml deployed in this lab, reflecting all five custom detection rules in their final working state:

```
<group name="custom_windows_hids,">

  <!-- Rule 1: Windows failed logon (Event ID 4625) -->
  <rule id="100001" level="6">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4625$</field>
    <description>Custom: Windows failed logon attempt (Event
4625)</description>
    <group>authentication_failed,pci_dss_10.2.4,</group>
  </rule>

  <!-- Rule 2: Brute force - 5 logon failures in 60 seconds -->
  <rule id="100002" level="12" frequency="5" timeframe="60">
    <if_matched_sid>100001</if_matched_sid>
    <same_source_ip />
    <description>Custom: Windows Brute Force - 5+ failures in 60s</description>
    <group>authentication_failures,pci_dss_10.2.4,gdpr_IV_35.7.d,</group>
  </rule>

  <!-- Rule 3: File modification detected via FIM (eventchannel 4663
limitation) -->
  <rule id="100003" level="8">
    <if_sid>550</if_sid>
    <description>Custom: Sensitive file modification detected via FIM
(WazuhTest)</description>
    <group>file_access,pci_dss_10.2.7,syscheck,</group>
  </rule>

  <!-- Rule 4: USB / Removable storage connected (Event 6416) -->
  <rule id="100004" level="7">
    <if_sid>60000</if_sid>
    <field name="win.system.eventID">^2003$</field>
    <description>Custom: Removable storage device connected</description>
    <group>removable_storage,</group>
  </rule>

  <!-- Rule 5: SSH Brute Force - Active Response trigger -->
  <rule id="100005" level="12" frequency="5" timeframe="60">
    <if_matched_sid>5712</if_matched_sid>
    <same_source_ip />
    <description>Custom: SSH Brute Force detected - triggering Active
Response</description>
    <group>authentication_failures,</group>
  </rule>

</group>
```

Appendix A — Full Command Reference

This appendix consolidates every command executed throughout the lab, in chronological order, covering Wazuh manager/indexer/dashboard deployment, certificate generation, Windows agent enrollment, troubleshooting, custom detection rules, audit policy configuration, attack simulation, and Filebeat-to-Indexer pipeline repair.

A.1 Kali — Wazuh Repository Setup

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o
/usr/share/keyrings/wazuh.gpg

echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg]
https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee
/etc/apt/sources.list.d/wazuh.list

sudo apt update
```

A.2 Kali — Install Wazuh Manager, Indexer, Dashboard

```
sudo apt install wazuh-manager wazuh-indexer wazuh-dashboard -y
```

A.3 Kali — Generate TLS Certificates

```
curl -sO https://packages.wazuh.com/4.14/wazuh-certs-tool.sh

cat > config.yml << 'EOF'
nodes:
  indexer:
    - name: node-1
      ip: 192.168.18.136
  server:
    - name: wazuh-1
      ip: 192.168.18.136
  dashboard:
    - name: dashboard
      ip: 192.168.18.136
EOF

sudo rm -rf wazuh-certificates wazuh-certificates.tar
sudo bash wazuh-certs-tool.sh -A
sudo ls -la wazuh-certificates/
```

A.4 Kali — Deploy Indexer Certificates

```
sudo mkdir -p /etc/wazuh-indexer/certs
sudo cp wazuh-certificates/node-1.pem /etc/wazuh-indexer/certs/indexer.pem
sudo cp wazuh-certificates/node-1-key.pem /etc/wazuh-indexer/certs/indexer-
key.pem
sudo cp wazuh-certificates/root-ca.pem /etc/wazuh-indexer/certs/root-ca.pem
sudo cp wazuh-certificates/admin.pem /etc/wazuh-indexer/certs/admin.pem
sudo cp wazuh-certificates/admin-key.pem /etc/wazuh-indexer/certs/admin-key.pem
sudo chmod 500 /etc/wazuh-indexer/certs
sudo chmod 400 /etc/wazuh-indexer/certs/*
sudo chown -R wazuh-indexer:wazuh-indexer /etc/wazuh-indexer/certs
```

A.5 Kali — Deploy Dashboard Certificates

```
sudo mkdir -p /etc/wazuh-dashboard/certs
sudo cp wazuh-certificates/dashboard.pem /etc/wazuh-
dashboard/certs/dashboard.pem
sudo cp wazuh-certificates/dashboard-key.pem /etc/wazuh-
dashboard/certs/dashboard-key.pem
sudo cp wazuh-certificates/root-ca.pem /etc/wazuh-dashboard/certs/root-ca.pem
sudo chmod 500 /etc/wazuh-dashboard/certs
sudo chmod 400 /etc/wazuh-dashboard/certs/*
sudo chown -R wazuh-dashboard:wazuh-dashboard /etc/wazuh-dashboard/certs
```

A.6 Kali — Enable & Start Core Services

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-indexer wazuh-manager wazuh-dashboard
sudo systemctl start wazuh-indexer wazuh-manager wazuh-dashboard
sudo systemctl status wazuh-indexer wazuh-manager wazuh-dashboard --no-pager
```

A.7 Kali — Initialize Indexer Security

```
sudo /usr/share/wazuh-indexer/bin/indexer-security-init.sh
```

A.8 Kali — Firewall Rules for Agent Communication

```
sudo ufw allow 1514/tcp
sudo ufw allow 1514/udp
sudo ufw allow 1515/tcp
```

A.9 Windows — Download & Install Wazuh Agent (CMD)

```
curl -o "%TEMP%\wazuh-agent.msi" "https://packages.wazuh.com/4.x/windows/wazuh-
agent-4.14.5-1.msi"

msiexec /i "%TEMP%\wazuh-agent.msi" /q WAZUH_MANAGER="192.168.18.136"

net start WazuhSvc
```

A.10 Windows — Fix Missing internal_options.conf Definitions (PowerShell)

The bundled internal_options.conf shipped incomplete with the 4.14.5 Windows MSI, causing the agent to crash on startup with 'Definition not found' errors. The fix downloads the complete reference file from the official Wazuh source and supplements local_internal_options.conf:

```
# Replace with the complete reference internal_options.conf
Invoke-WebRequest -Uri
"https://raw.githubusercontent.com/wazuh/wazuh/v4.14.5/etc/internal_options.con
f" -OutFile "C:\Program Files (x86)\ossec-agent\internal_options.conf"

# Supplement local overrides (first batch)
```

```
$options = @"
agent.recv_timeout=60
agent.remote_conf=1
agent.min_eps=50
agent.max_eps=1000
agent.eps_limit=0
agent.warn_level=7
agent.debug_level=0
"@
Add-Content "C:\Program Files (x86)\ossec-agent\local_internal_options.conf"
$options
```

A.11 Windows — Fix Manager IP & Client Block in ossec.conf

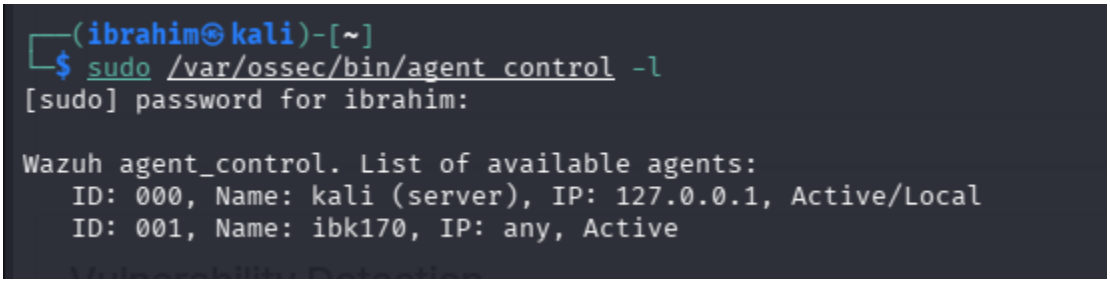
```
net stop WazuhSvc
net start WazuhSvc
```

Manual edit to the bottom of ossec.conf (replacing the legacy <server-ip> tag with the modern <server> block):

```
<ossec_config>
  <client>
    <server>
      <address>192.168.18.136</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
  </client>
</ossec_config>
```

A.12 Kali — Verify Agent Connection

```
sudo /var/ossec/bin/agent_control -l
```



A.13 Kali — Static IP Configuration (preventing DHCP drift)

Mid-lab, the Kali VM's DHCP-assigned IP changed from 192.168.18.135 to 192.168.18.136, breaking agent connectivity. This was resolved by updating the agent and dashboard configs, and recommending a static IP assignment going forward:

```
ip a

sudo nmcli con mod "Wired connection 1" ipv4.addresses 192.168.18.136/24
ipv4.gateway 192.168.18.1 ipv4.dns 8.8.8.8 ipv4.method manual
sudo nmcli con up "Wired connection 1"
```

A.14 Kali — Custom Detection Rules (local_rules.xml)

```
sudo mkdir -p /var/ossec/rules
sudo nano /var/ossec/rules/local_rules.xml


<group name="custom_windows_hids,">

  <!-- Rule 1: Windows failed logon (Event ID 4625) -->
  <rule id="100001" level="6">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4625$</field>
    <description>Custom: Windows failed logon attempt (Event
4625)</description>
    <group>authentication_failed,pci_dss_10.2.4,</group>
  </rule>

  <!-- Rule 2: Brute force - 5 logon failures in 60 seconds -->
  <rule id="100002" level="12" frequency="5" timeframe="60">
    <if_matched_sid>100001</if_matched_sid>
    <same_source_ip />
    <description>Custom: Windows Brute Force - 5+ failures in 60s</description>
    <group>authentication_failures,pci_dss_10.2.4,gdpr_IV_35.7.d,</group>
  </rule>

  <!-- Rule 3: File modification detected via FIM (eventchannel 4663
limitation) -->
  <rule id="100003" level="8">
    <if_sid>550</if_sid>
    <description>Custom: Sensitive file modification detected via FIM
(WazuhTest)</description>
    <group>file_access,pci_dss_10.2.7,syscheck,</group>
  </rule>

  <!-- Rule 4: USB / Removable storage connected (Event 6416) -->
  <rule id="100004" level="7">
    <if_sid>60000</if_sid>
    <field name="win.system.eventID">^2003$</field>
    <description>Custom: Removable storage device connected</description>
    <group>removable_storage,</group>
  </rule>

  <!-- Rule 5: SSH Brute Force - Active Response trigger -->
  <rule id="100005" level="12" frequency="5" timeframe="60">
    <if_matched_sid>5712</if_matched_sid>
    <same_source_ip />
    <description>Custom: SSH Brute Force detected - triggering Active
Response</description>
    <group>authentication_failures,</group>
  </rule>

</group>
```

```
# Validate rule syntax before restart
sudo /var/ossec/bin/wazuh-analysisd -t

sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager --no-pager | grep -E "Active|running"
```

A.15 Windows — Enable Audit Policies

```
auditpol /set /category:"Object Access" /success:enable /failure:enable
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
auditpol /set /subcategory:"Credential Validation" /success:enable
/failure:enable
auditpol /set /subcategory:"User Account Management" /success:enable
/failure:enable
auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
auditpol /set /subcategory:"Audit Policy Change" /success:enable
/failure:enable
auditpol /set /subcategory:"Removable Storage" /success:enable /failure:enable

# Verify
auditpol /get /category:"Object Access"
auditpol /get /category:"Logon/Logoff","Detailed Tracking","Account Management"
```

A.16 Windows — Brute Force / Failed Logon Simulation

```
1..5 | ForEach-Object {
  $pass = ConvertTo-SecureString "wrongpass123" -AsPlainText -Force
  $cred = New-Object System.Management.Automation.PSCredential("fakeuser",
$pass)
  Start-Process cmd -Credential $cred -ErrorAction SilentlyContinue
  Start-Sleep -Seconds 1
}
```

```
# Verify Event ID 4625 entries generated
Get-WinEvent -LogName Security -FilterXPath "*[System[EventID=4625]]" -
MaxEvents 10 | Select-Object TimeCreated, Message
```

A.17 Kali — Live Alert Verification

```
sudo tail -f /var/ossec/logs/alerts/alerts.log | grep -i "4625|brute|failed"

sudo grep -i "4625|100001|100002|failed logon"
/var/ossec/logs/alerts/alerts.log | tail -20
```

A.18 Kali — Wazuh API Password Reset

During verification, default API/dashboard credentials failed authentication. The Wazuh password reset utility was used to regenerate strong indexer credentials:

```
curl -sO https://packages.wazuh.com/4.14/wazuh-passwords-tool.sh
sudo bash wazuh-passwords-tool.sh -a
```

```
# Test API authentication with the working wazuh-wui account
curl -k -u wazuh-wui:wazuh-wui -X POST
https://localhost:55000/security/user/authenticate
```

A.19 Kali — Filebeat Installation & Indexer Pipeline Repair

The dashboard initially showed 'No results' for Threat Hunting despite alerts.log containing valid entries. Root cause: Filebeat (the shipper that forwards alerts from the manager to the indexer) was not installed. The default apt package also proved version-incompatible and was replaced with the Wazuh-distributed OSS build:

```
# Remove incompatible default filebeat
sudo systemctl stop filebeat
sudo apt remove filebeat -y

# Install Wazuh-compatible filebeat OSS build
curl -sO https://packages.wazuh.com/4.x/apt/pool/main/f/filebeat/filebeat-oss-
7.10.2-amd64.deb
sudo dpkg -i filebeat-oss-7.10.2-amd64.deb

# Apply Wazuh filebeat config and credentials
sudo curl -so /etc/filebeat/filebeat.yml
https://packages.wazuh.com/4.14/tpl/wazuh/filebeat/filebeat.yml
sudo sed -i 's|${password}|<INDEXER_ADMIN_PASSWORD>|g'
/etc/filebeat/filebeat.yml
sudo sed -i 's|${username}|admin|g' /etc/filebeat/filebeat.yml

# Wazuh index template and ingest module
sudo curl -so /etc/filebeat/wazuh-template.json
https://raw.githubusercontent.com/wazuh/wazuh/v4.14.5/extensions/elasticsearch/
7.x/wazuh-template.json
sudo curl -s https://packages.wazuh.com/4.x/filebeat/wazuh-filebeat-0.4.tar.gz
| sudo tar -xvz -C /usr/share/filebeat/module

# Filebeat certificates
sudo mkdir -p /etc/filebeat/certs
sudo cp wazuh-certificates/root-ca.pem /etc/filebeat/certs/root-ca.pem
sudo cp wazuh-certificates/wazuh-1.pem /etc/filebeat/certs/filebeat.pem
sudo cp wazuh-certificates/wazuh-1-key.pem /etc/filebeat/certs/filebeat-key.pem
sudo chmod 500 /etc/filebeat/certs
sudo chmod 400 /etc/filebeat/certs/*

# Clear stale registry causing crawler crash, then start
sudo rm -rf /var/lib/filebeat/registry
sudo systemctl enable filebeat
sudo systemctl start filebeat
sudo systemctl status filebeat --no-pager | grep Active
```

A.20 Kali — Confirm Alerts are Indexed

```
curl -k -u admin:"<INDEXER_ADMIN_PASSWORD>"
https://localhost:9200/_cat/indices?v | grep wazuh-alerts
```

Result: wazuh-alerts-4.x-2026.06.29 index created, green health, 2099+ documents - confirming the full pipeline (Windows Agent → Manager → Filebeat → Indexer → Dashboard) was operating correctly end-to-end.

A.21 Final Full-Stack Verification

```
# All services running
sudo systemctl status wazuh-indexer wazuh-manager wazuh-dashboard filebeat --
no-pager | grep -E "Active"

# Agent connected
sudo /var/ossec/bin/agent_control -l

# Alerts indexed
curl -k -u admin:"<INDEXER_ADMIN_PASSWORD>"
https://localhost:9200/_cat/indices?v | grep wazuh-alerts

# API responding
curl -k -u wazuh-wui:wazuh-wui -X POST
https://localhost:55000/security/user/authenticate

# Manager daemon status
sudo /var/ossec/bin/wazuh-control status
```

References

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